

Problem 1

11.1. Consider the interconnection of discrete-time LTI systems shown in Figure P11.1. Express the overall system function for this interconnection in terms of $H_0(z)$, $H_1(z)$, and $G(z)$.

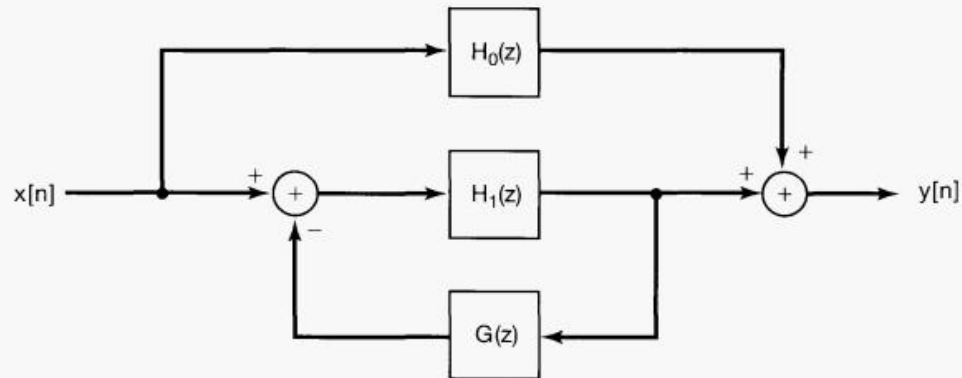
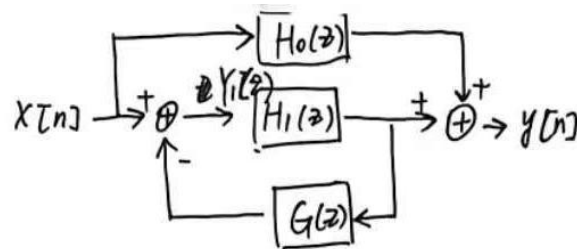


Figure P11.1

Problem 1



$$Y_1(z) = X(z) - G(z)H_1(z) \cdot Y_1(z) \Rightarrow Y_1(z) = \frac{X(z)}{1 + G(z)H_1(z)}$$

$$Y(z) = Y_1(z) \cdot H_1(z) + X(z) \cdot H_0(z)$$

$$= \left(\frac{H_1(z)}{1 + H_1(z)G(z)} + H_0(z) \right) X(z)$$

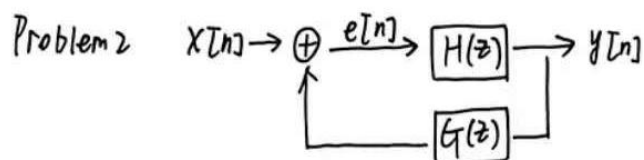
$$\Rightarrow H(z) = H_0(z) + \frac{H_1(z)}{1 + H_1(z)G(z)}$$

Problem 2

11.5. Consider the discrete-time feedback system depicted in Figure 11.3(b) with

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \quad \text{and} \quad G(z) = 1 - bz^{-1}.$$

For what real values of b is the feedback system stable?



把 $H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$, $G(z) = 1 - \frac{b}{z}$ 代入 $Q(z) = \frac{H(z)}{1 + H(z)G(z)}$

$$\Rightarrow Q(z) = \frac{1}{z - \frac{0.5+b}{2}} \quad \text{极点: } z = \frac{0.5+b}{2}$$

稳定 $\Rightarrow |z| < 1 \Rightarrow \left| \frac{0.5+b}{2} \right| < 1 \Rightarrow b \in (-2.5, 1.5)$

Problem 3

11.7. Suppose the closed-loop poles of a feedback system satisfy

$$\frac{1}{(s+2)(s+3)} = -\frac{1}{K}.$$

Use the root-locus method to determine the values of K for which the feedback system is guaranteed to be stable.

Problem 3

$$\frac{1}{(s+2)(s+3)} = -\frac{1}{K} \Rightarrow s^2 + 5s + 6 + K = 0 \Rightarrow s^2 + 5s + 6 = -K$$

解出两 s 均在左平面:  $\Rightarrow -K < 0 \Rightarrow K > 0$

Problem 4

11.57. In this problem, we consider several examples of the design of tracking systems. Consider the system depicted in Figure P11.57. Here, $H_p(s)$ is the system whose output is to be controlled, and $H_c(s)$ is the compensator to be designed. Our objective in choosing $H_c(s)$ is that we would like the output $y(t)$ to follow the input $x(t)$. In particular, in addition to stabilizing the system, we would also like to design the system so that the error $e(t)$ decays to zero for certain specified inputs.

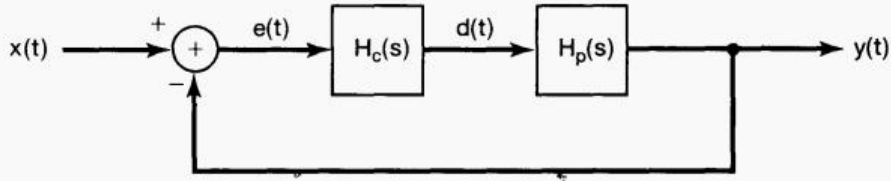


Figure P11.57

(a) Suppose that

$$H_p(s) = \frac{\alpha}{s + \alpha}, \quad \alpha \neq 0. \quad (\text{P11.57-1})$$

Show that if $H_c(s) = K$ (which is known as *proportional* or *P* control), we can choose K so as to stabilize the system and so that $e(t) \rightarrow 0$ if $x(t) = \delta(t)$. Show that we *cannot* get $e(t) \rightarrow 0$ if $x(t) = u(t)$.

(b) Again let $H_p(s)$ be as in eq. (P11.57-1), and suppose that we use *proportional-plus-integral* (PI) control—that is,

$$H_c(s) = K_1 + \frac{K_2}{s}.$$

Show that we can choose K_1 and K_2 so as to stabilize the system, and we can also get $e(t) \rightarrow 0$ if $x(t) = u(t)$. Thus, the system can track a step. In fact, this illustrates a basic and important principle in feedback system design: To track a step [$X(s) = 1/s$], we need an integrator ($1/s$) in the feedback system. An extension of this principle is considered in the next problem.

(c) Suppose that

$$H_p(s) = \frac{1}{(s-1)^2}.$$

Show that we cannot stabilize this system with a PI controller, but that we can stabilize it and have it track a step if we use *proportional-plus-integral-plus-differential* (PID) control, i.e.,

$$H_c(s) = K_1 + \frac{K_2}{s} + K_3 s.$$

Problem 4. $E(s) = X(s) - Y(s) = X(s) - H_c(s) H_p(s) E(s)$

$$\Rightarrow E(s) = \frac{X(s)}{1 + H_c(s) H_p(s)} \quad \text{稳定: 极点在左半平面.}$$

(a) $H_p(s) = \frac{2}{s+2}$ $H_c(s) = K$,

$$1 + K \cdot \frac{2}{s+2} = 0 \Rightarrow s = -2(1+K) \Rightarrow \text{稳定: } 2(1+K) > 0$$

① $x(t) = \delta(t)$ 时, $X(s) = 1$

$$E(s) = \frac{1}{1 + K \cdot \frac{2}{s+2}} = \frac{s+2}{s+2(1+K)}$$

$$e(\infty) = \lim_{s \rightarrow 0} s E(s) = 0. \quad \text{可以使 } e(t) \rightarrow 0.$$

② $x(t) = u(t)$ 时, $X(s) = \frac{1}{s}$.

$$E(s) = \frac{1}{s(1 + K \frac{2}{s+2})} \Rightarrow e(\infty) = \lim_{s \rightarrow 0} \frac{1}{1 + K \frac{2}{s+2}} = \frac{1}{1+K} \quad K \text{ 有限} \Rightarrow e(t) \text{ 不} \rightarrow 0.$$

(b) 已知 $H_p(s) = \frac{2}{s+2}$, $H_c(s) = K_1 + \frac{K_2}{s}$

$$X(s) = \frac{1}{s} \text{ 时, } E(s) = \frac{1}{1 + \frac{2}{s+2} \cdot (K_1 + \frac{K_2}{s})} \cdot \frac{1}{s}$$

$$e(\infty) = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} \frac{1}{1 + \frac{2}{s+2} (K_1 + \frac{K_2}{s})} = 0.$$

$\Rightarrow$ 引入积分环节后系统可以跟踪阶跃输入, 使 $e(t) \rightarrow 0$.

分析稳定性:

$$1 + \frac{2}{s+2} \cdot (K_1 + \frac{K_2}{s}) = 0$$

$$\Rightarrow s^2 + 2(1+K_1)s + 2K_2 = 0$$

时二阶系统, 所有系数同号, 系统就稳定

$$\begin{cases} 2(1+K_1) > 0 \\ 2K_2 > 0 \end{cases}$$

(c) 已知 $H_p(s) = \frac{1}{(s-1)^2}$

① 证明无法用PI控制器使其稳定.

$$\text{令 } H_c(s) = K_1 + \frac{K_2}{s}, \quad 1 + \frac{1}{(s-1)^2} \cdot (K_1 + \frac{K_2}{s}) = 0$$

$$\Rightarrow s^3 - 2s^2 + (1+K_1)s + K_2 = 0.$$

不同号. 无法做到所有根在左半平面. $\Rightarrow$ 不能稳定.

$$\begin{cases} \text{积分} \rightarrow \frac{K_i}{s} \\ \text{微分} \rightarrow K_d \cdot s \end{cases}$$

② 证明PID控制器可以稳定.

对 $a_n s^n + \dots + a_1 s + a_0 = 0$, 所有解在右半平面:

$a_n \sim a_0$ 符号相同.

① 若 $a_i = 0$ / 符号不同 $\Rightarrow$ 一定不稳定. ② 符号相同 $\begin{cases} \text{二阶: 稳定} \\ \text{多阶: 劳斯表.} \end{cases}$

$$\text{令 } H_c(s) = K_1 + \frac{K_2}{s} + K_3 s.$$

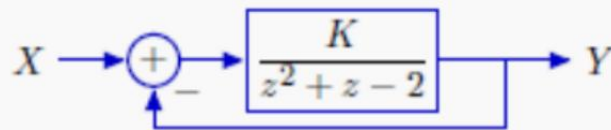
$$1 + \frac{1}{(s-1)^2} (K_1 + \frac{K_2}{s} + K_3 s) = 0 \Rightarrow s^3 + (K_3 - 2)s^2 + (K_1 + 1)s + K_2 = 0$$

$$\text{取 } \begin{cases} K_3 > 2, K_1 > -1, K_2 > 0 \\ (K_3 - 2)(K_1 + 1) > K_2 \end{cases} \text{ 即使系统稳定.}$$

$$\text{eg: 三阶 } a_3 s^3 + a_2 s^2 + a_1 s + a_0 = 0 \\ a_2 a_1 > a_3 a_0 \text{ 即稳定.}$$

Problem 5

Consider the following feedback system in which the box represents a causal LTI DT system that is represented by its system function.



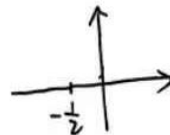
- Determine the range of K for which this feedback system is stable.
- Determine the range of K for which this feedback system has real-valued poles.

Problem 5

$$Q(z) = \frac{\frac{K}{z^2+z-2}}{1 + \frac{K}{z^2+z-2}} = \frac{K}{z^2+z-2+K}$$

$$z^2+z-2+K=0$$

(a) 系统稳定: $K > 0 \Rightarrow K > 2$.



解得 $|z| < 1$. 不是在右半平面!

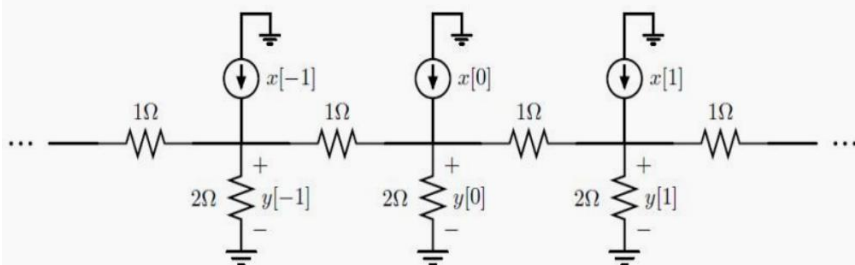
$$z_0 = -\frac{1}{2} \pm \frac{1}{2} \sqrt{9-4K}, \text{ 临界 } \sqrt{9-4K} = 7 \text{ 时, } K=3. \Rightarrow K \in (2, 3)$$

(b) 系统有实数极点:

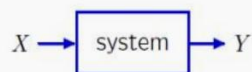
$$9-4K \geq 0 \Rightarrow K \leq \frac{9}{4}$$

Problem 6

An infinite network of resistors is excited by an infinite network of current sources as shown below.

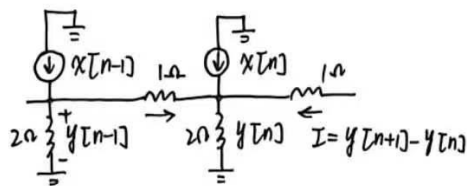


We can consider the transformation from x to y as a DT system.



- Show that this system is linear and “time”-invariant.
- Determine the unit-sample response $h[n]$.
- Determine the system function $H(z)$ and region of convergence.
- Determine the system’s pole(s) and zero(s).

Problem b.



$$x[n] + y[n+1] - y[n] + y[n-1] - y[n] = \frac{1}{2}y[n].$$

$$\Rightarrow -y[n-1] + 2.5y[n] - y[n+1] = x[n]$$

(a) 证明系统线性时不变。由推导出的差分方程可知。

(b) 求 $h[n]$ 。

$$-\frac{1}{2}Y(z) + 2.5Y(z) - zY(z) = X(z)$$

$$\Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{z}{-z^2 + 2.5z - 1}$$

$$\text{写成 } \frac{z}{z-2} \text{ 形式: } H(z) = \frac{2}{3} \frac{z}{z-0.5} - \frac{2}{3} \frac{z}{z-2}$$

$$\Rightarrow h[n] = \frac{2}{3} (0.5)^n u[n] + \frac{2}{3} (2)^n u[-n-1] = \frac{2}{3} (0.5)^{|n|}$$

(c) 求 $H(z)$, ROC.

$$H(z) = \frac{z}{-z^2 + 2.5z - 1}, \text{ 极点 } 2, 0.5.$$

电压不可能无穷 $\Rightarrow$ 稳定 $\Rightarrow$ 包含 $|z|=1 \Rightarrow$ ROC: $0.5 < |z| < 2$.

(d) 确定极点零点:
 极点: $z=2, z=0.5$
 零点: $z=0$.